

THE MANY FORMS OF ENERGY

Tracking the Universe's Currency:
A structural blueprint to how physical systems
store, spend, and transform energy.

$$\text{STORED POTENTIAL} = \frac{k\Delta x^2}{2}$$

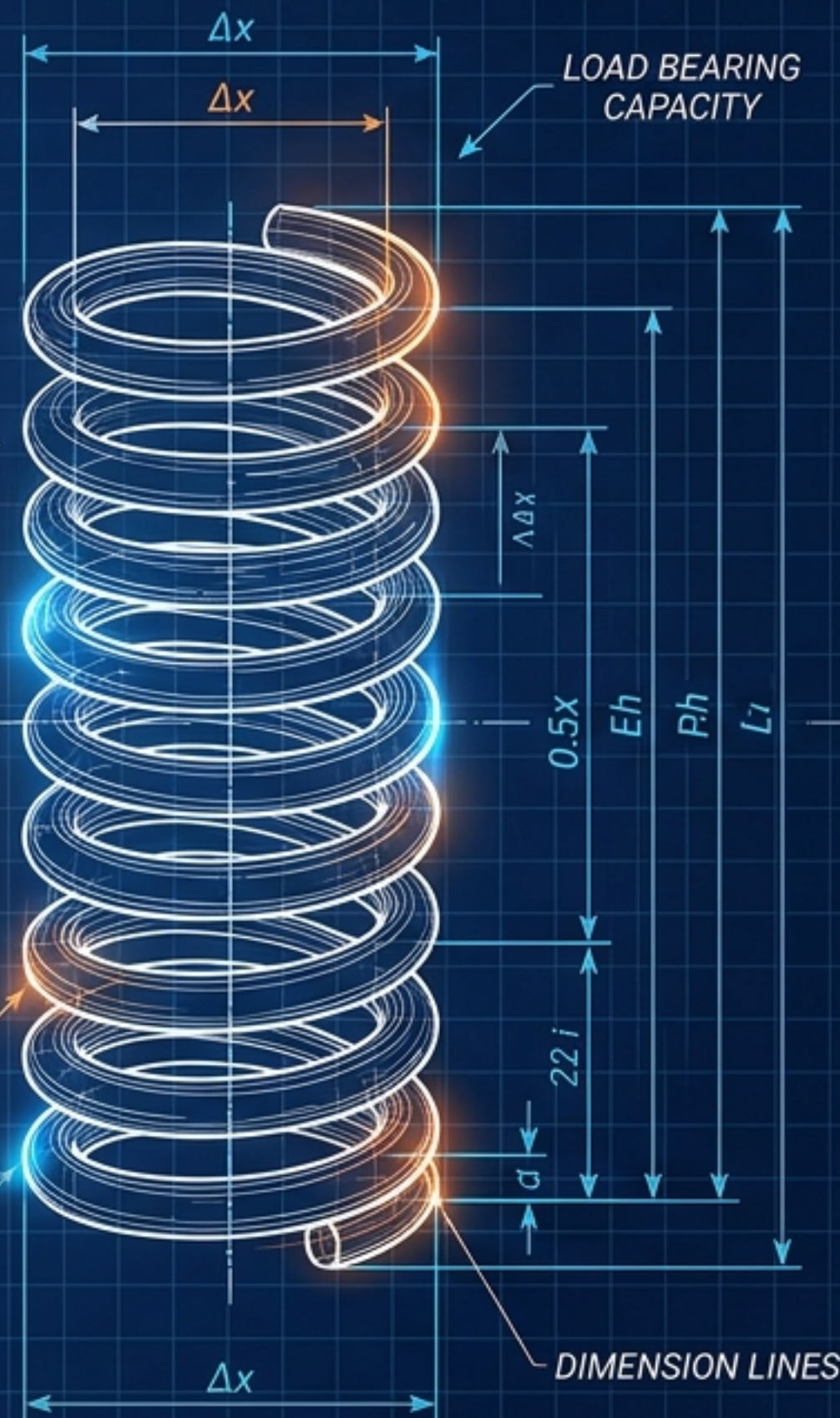
LOAD BEARING
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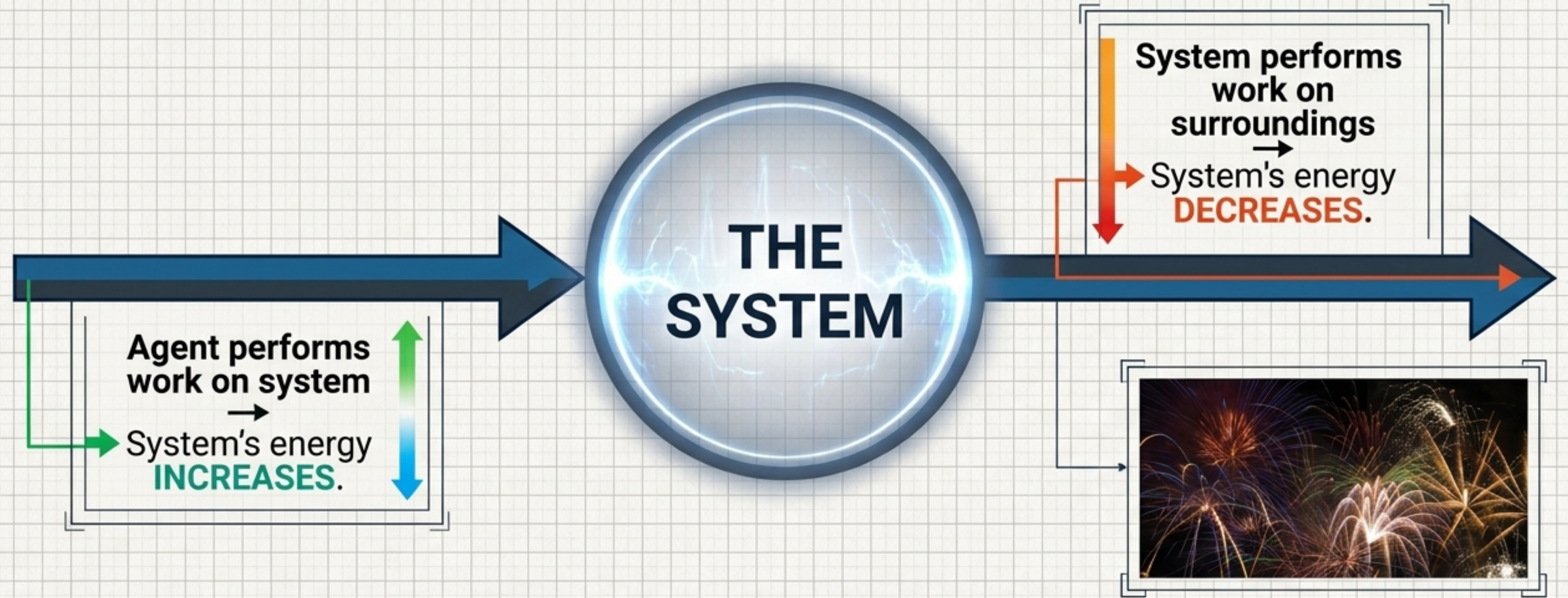
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STRUCTURAL
INTEGRITY

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DIMENSION LINES

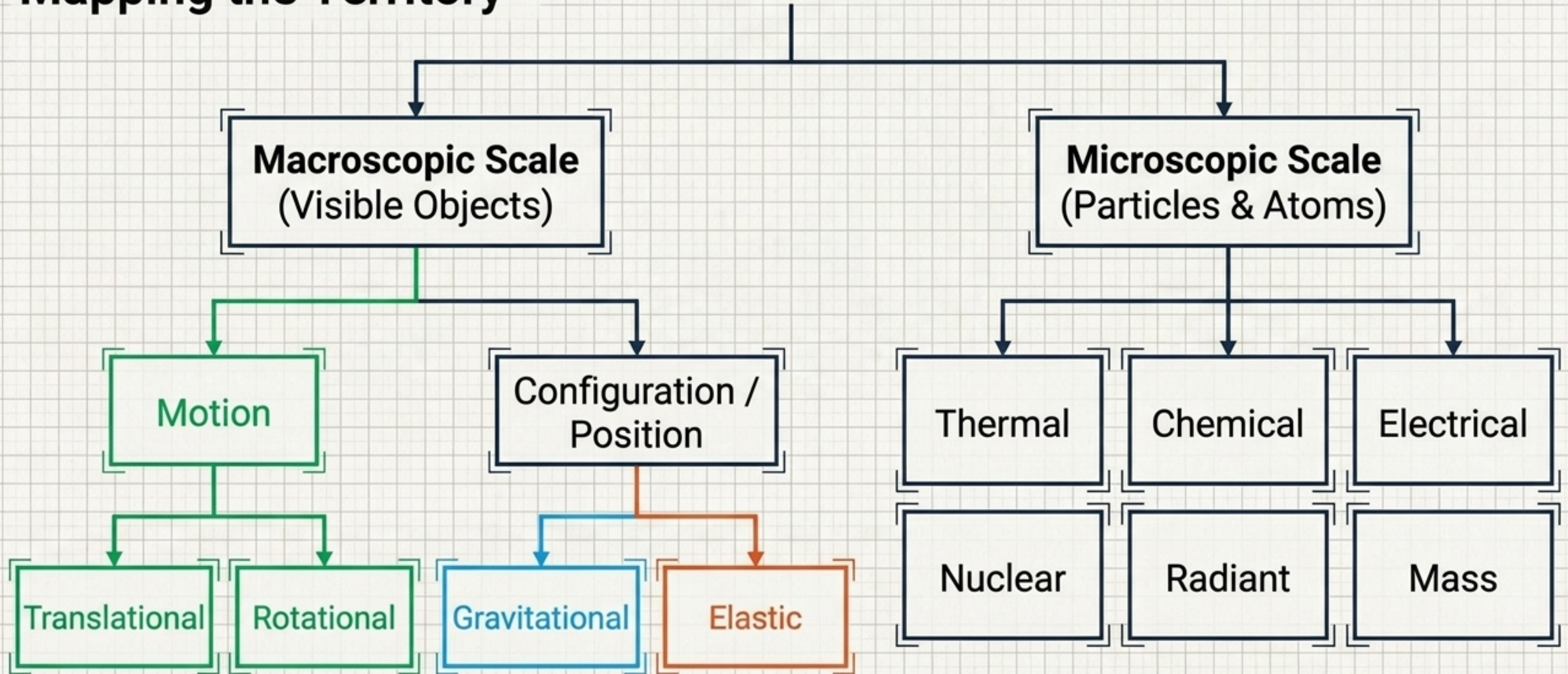




The Work-Energy Theorem:
Energy is the currency. Work is the transaction.

The Energy Taxonomy: Mapping the Territory

ENERGY



Kinetic Energy: The Currency of Motion

$$KE_{trans} = \frac{1}{2}mv^2$$



7.26-kg



0.148-kg

**Proportional to
MASS**
(Linear Scaling)
A bowling ball has
much more KE than a
baseball moving at
the identical speed.

**Proportional to
SPEED SQUARED**
(Exponential Scaling)
A car moving at 20
m/s has FOUR TIMES
the kinetic energy of a
car moving at 10 m/s.



Spinning Systems: Rotational Kinetic Energy



Calculation Key

$$KE_{rot} = \frac{1}{2} I \omega^2$$

I (Moment of Inertia) - Depends on the object's mass distribution.

omega (Angular Velocity) - The rate of spin.

An object can possess massive amounts of kinetic energy without changing its overall position in space. To make a top rotate, work must be done on it, transferring energy currency directly into its spin.

Translational vs. Rotational Kinetic Energy

Translational
(Point A to Point B)

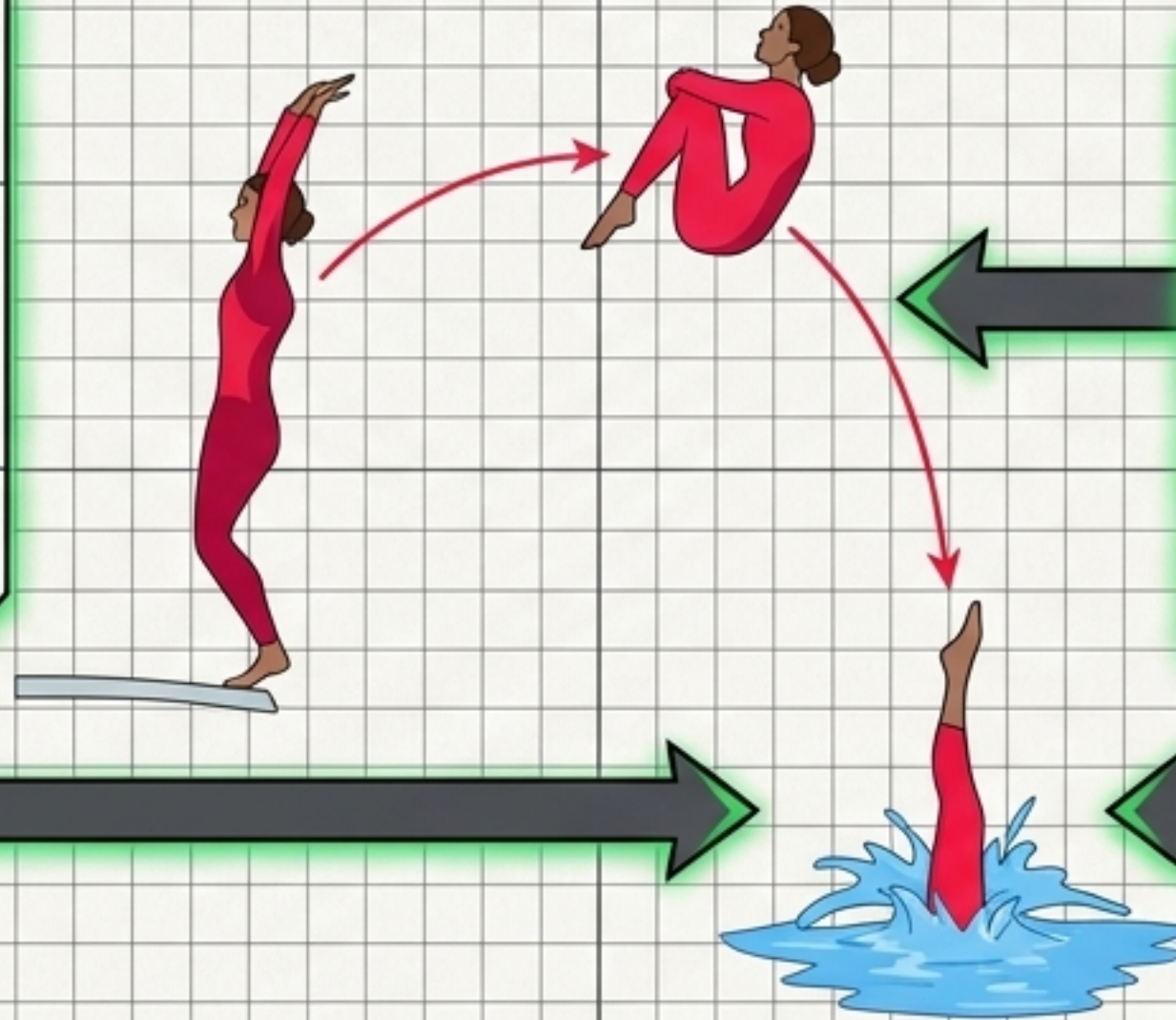
$$KE_{trans} = \frac{1}{2}mv^2$$

Depends on:
Total Mass

Rotational
(Spinning in Place)

$$KE_{rot} = \frac{1}{2}I\omega^2$$

Depends on:
Mass Distribution (Inertia)



The diver's center of mass moves
downward toward the pool.

The diver rotates rapidly around
her own center of mass.

The diving board does work on the diver, transferring **BOTH** types of kinetic energy simultaneously.

Potential Energy: Storing the Currency

The Metaphor



The Metaphor: Form changes. Total value remains exactly \$5.00.

The Physics

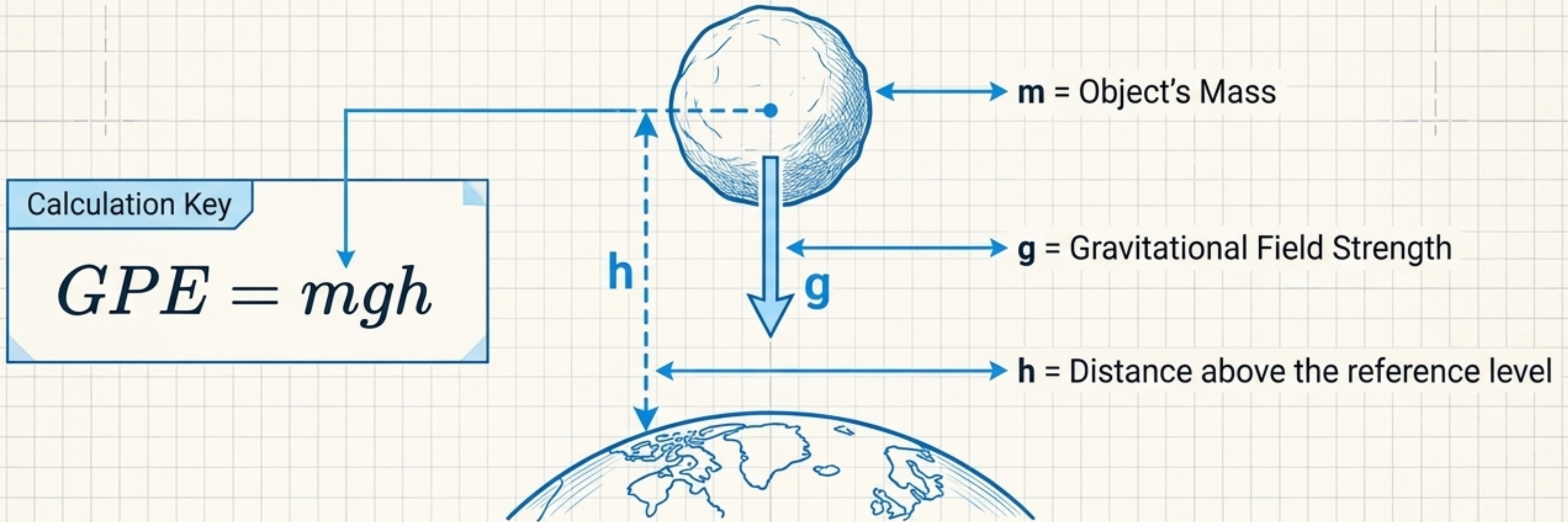


Active Currency:
Kinetic Energy

Stored Currency:
Potential Energy

The Physics: Energy is stored due to the interactions between objects in a system. It is the exact same currency, held in a different physical bank account waiting to be spent.

Gravitational Potential Energy (GPE)



Work Dynamics

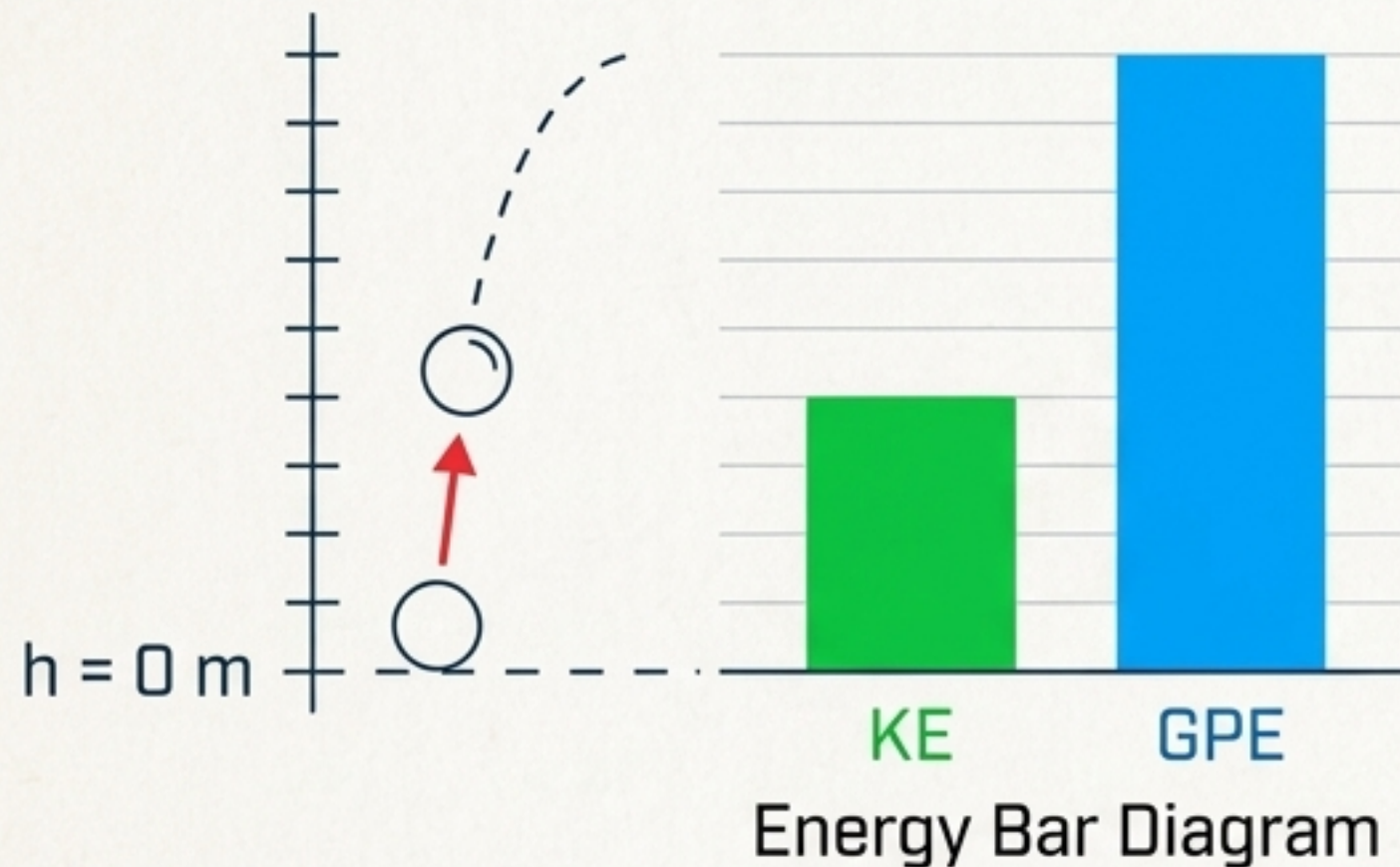
On the way down: Gravity does positive work (increases speed / KE).

On the way up: Gravity does negative work (decreases speed / KE).

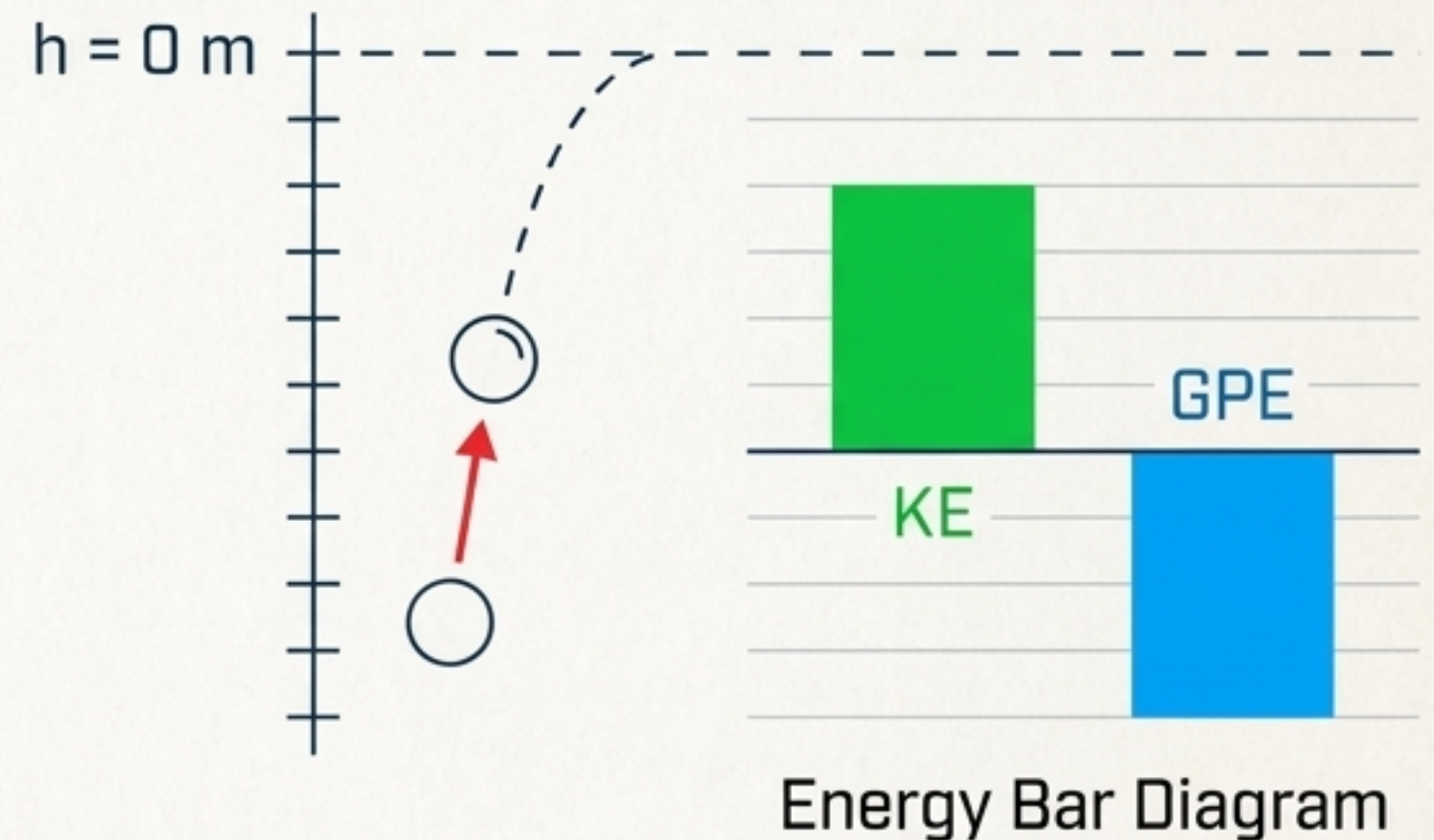
Note: The system requires BOTH the object and the Earth to exist.

The Power of Reference Levels

A Reference Level at Juggler's Hand



B Reference Level at Highest Point



Key Takeaway: The mathematical value of GPE changes based on where you draw the zero line, but the total energy of the system and the physical reality of the ball's flight remain exactly constant. It is merely a mathematical choice.

Elastic Potential Energy (EPE)



System: Bow, Arrow, Earth.
Work is done on the bow string, transferring energy to the severely bent limbs, storing Elastic Potential Energy.

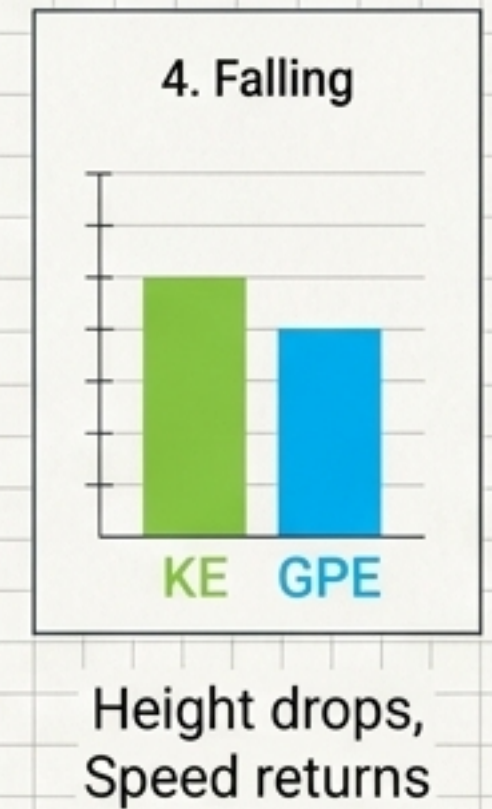
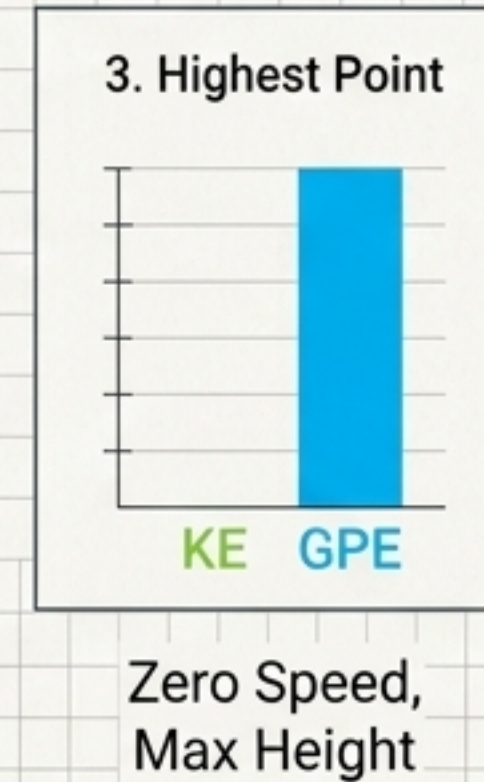
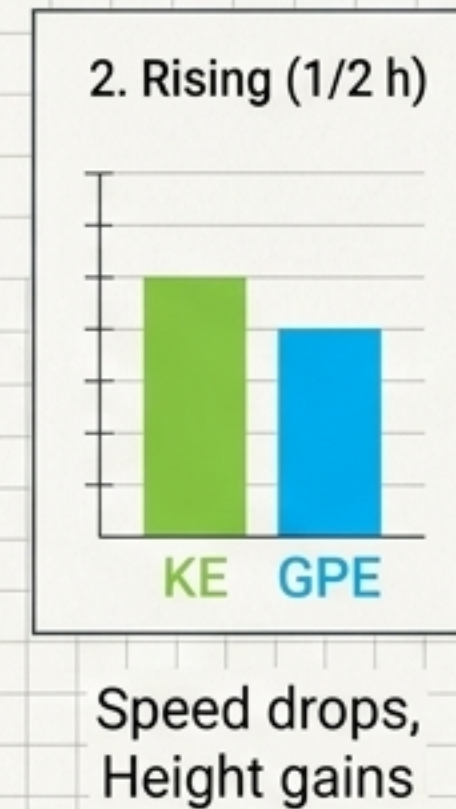
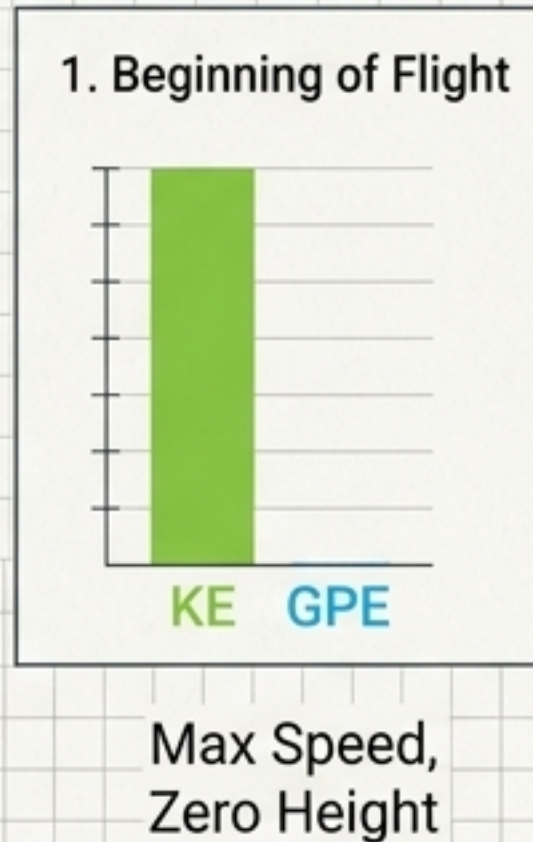
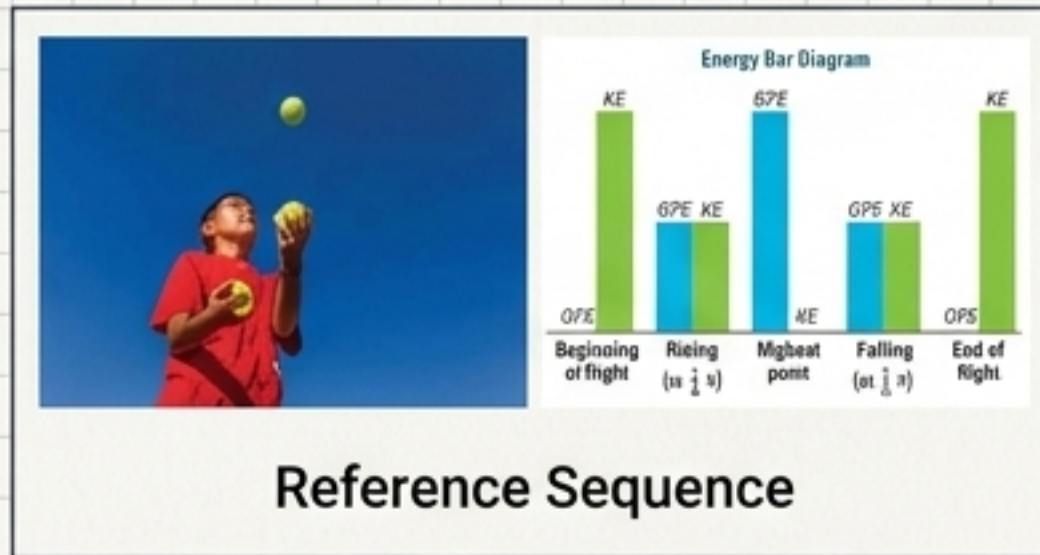


System: Vaulter, Pole, Earth.
The vaulter's kinetic energy forces the pole to bend, storing vast amounts of EPE.

Historical Tech Note

The Material Leap: Stiff bamboo poles stored almost no EPE because they didn't deform. The invention of flexible fiberglass allowed poles to bend dramatically, acting as massive energy batteries that revolutionized vaulting heights.

The Transformation Engine



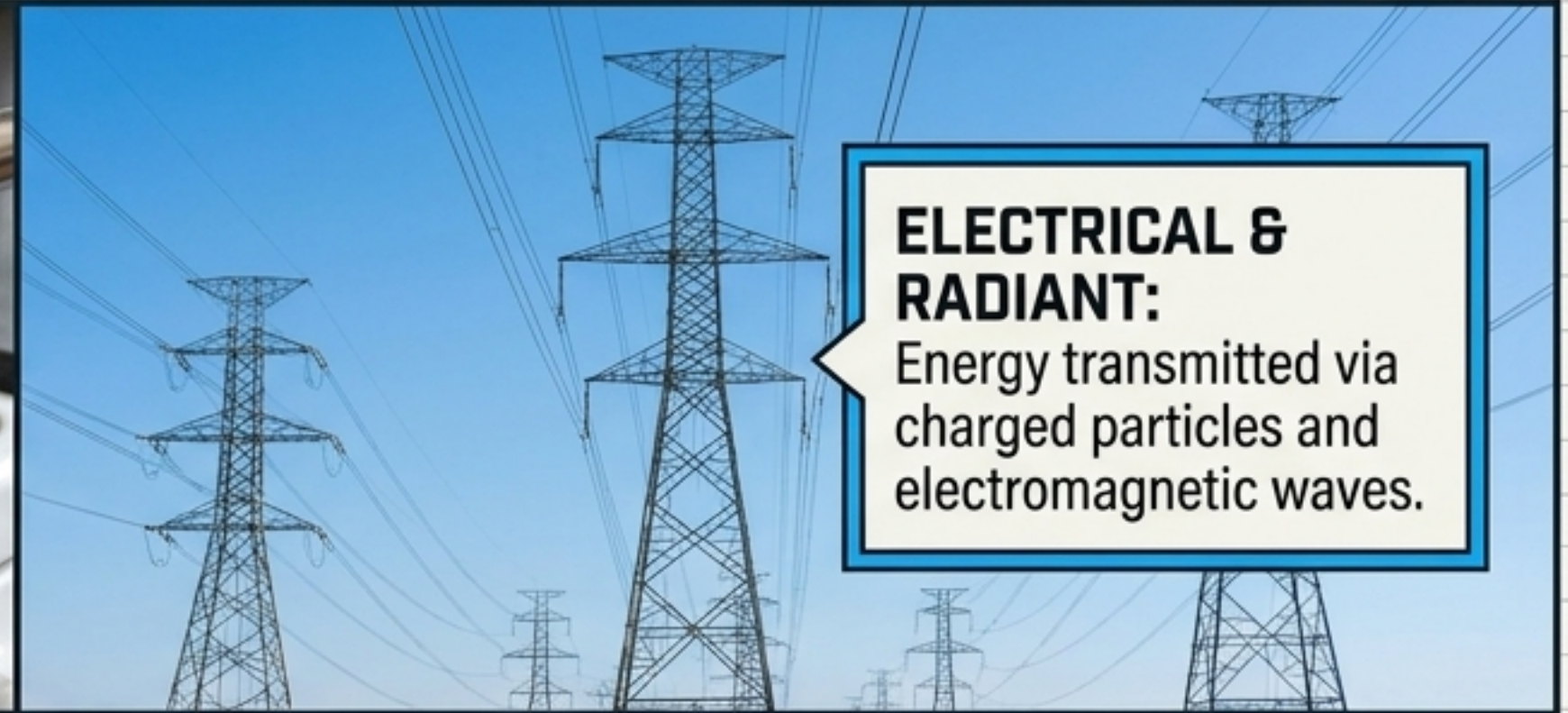
The proportions continuously invert, but the total height of the bars combined never changes. As one form depletes, the other perfectly fills up.

Microscopic & Subatomic Energy Forms



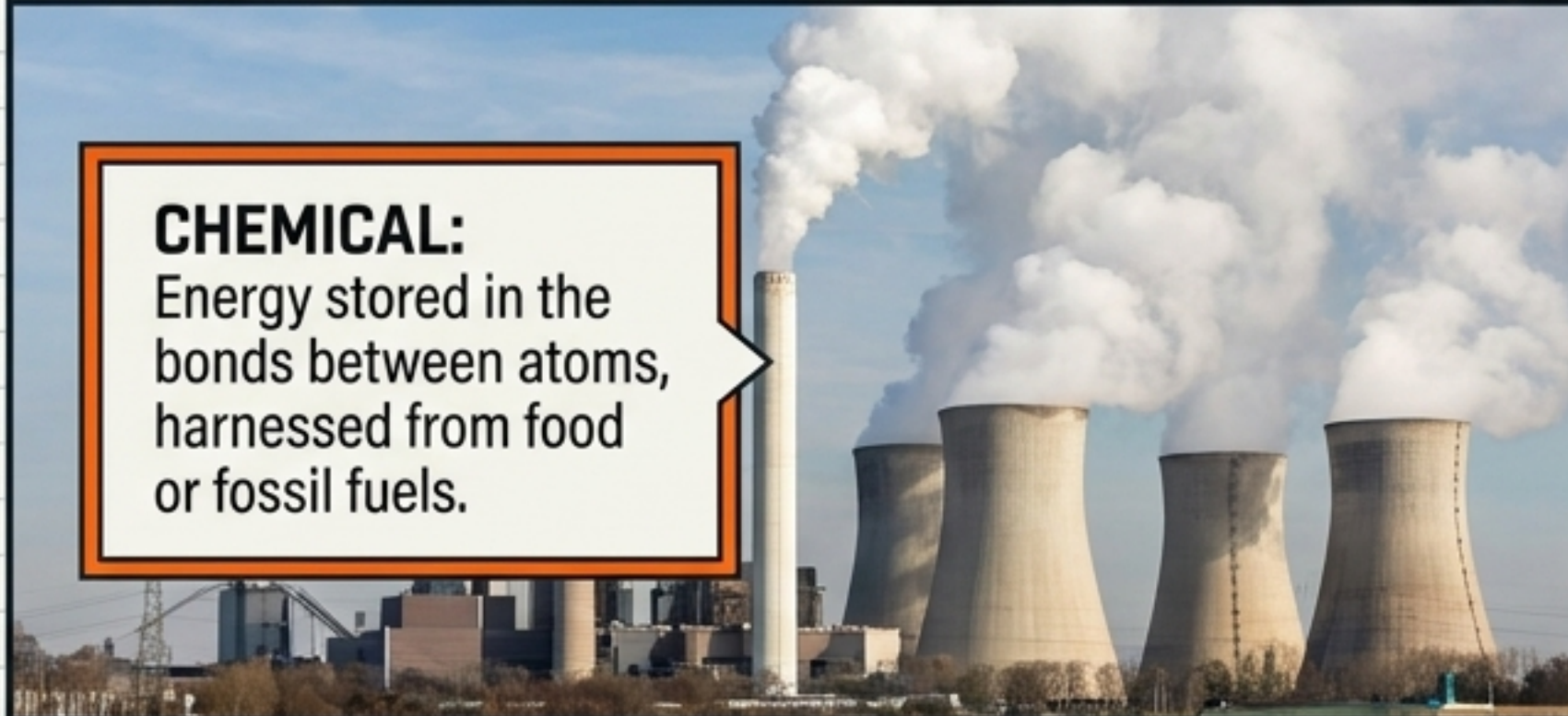
THERMAL:

The sum of kinetic and potential energy of moving microscopic particles.



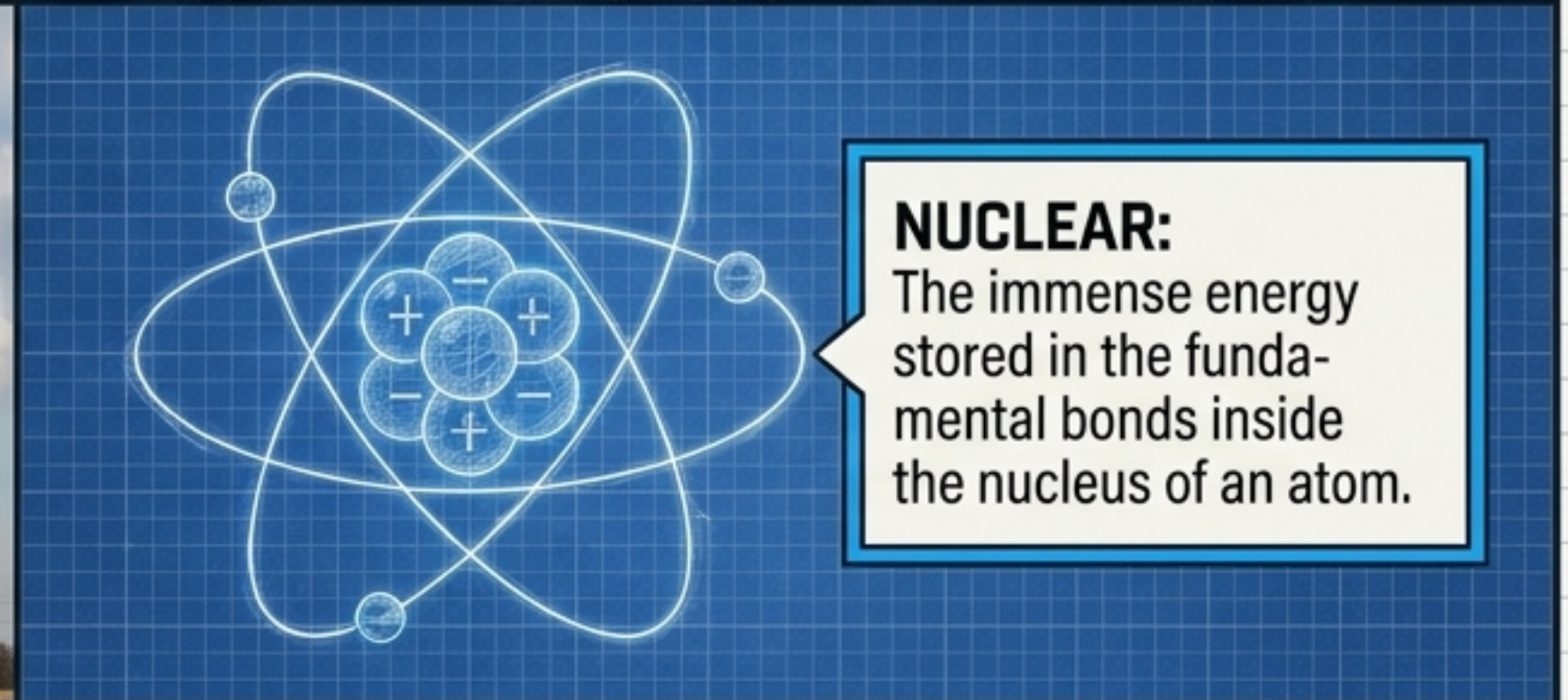
ELECTRICAL & RADIANT:

Energy transmitted via charged particles and electromagnetic waves.



CHEMICAL:

Energy stored in the bonds between atoms, harnessed from food or fossil fuels.



NUCLEAR:

The immense energy stored in the fundamental bonds inside the nucleus of an atom.

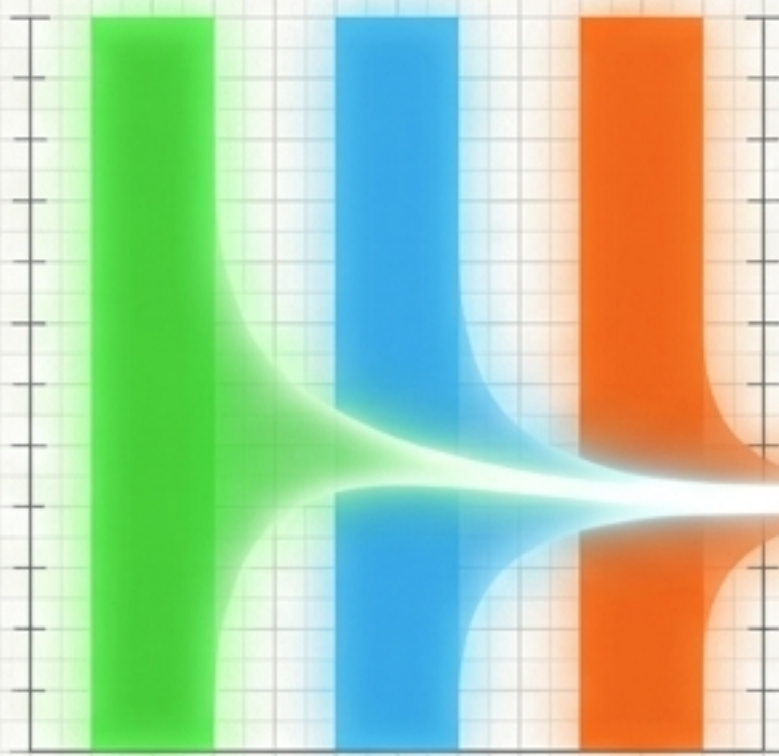
MASS: THE ULTIMATE EQUIVALENCE

$$E_0 = mc^2$$

Albert Einstein demonstrated that mass itself is a highly concentrated form of energy, known as rest energy. This profound mass-energy equivalence only becomes visually apparent when massive amounts of energy are involved, such as in nuclear explosions or particle physics experiments. Mass is not merely an object; it is frozen energy.

| The Energy Master Matrix

Energy Type	Scale	Key Formula	Trigger/Indicator
<u>Translational KE</u>	Macroscopic	$\frac{1}{2}mv^2$	Velocity
<u>Rotational KE</u>	Macroscopic	$\frac{1}{2}I\omega^2$	Spin / Angular Velocity
<u>Gravitational PE</u>	Macroscopic	mgh	Height / Distance between masses
<u>Elastic PE</u>	Macroscopic	(Depends on material)	Deformation / Shape Change
<u>Thermal Energy</u>	Microscopic	N/A	Temperature / Particle Motion
<u>Rest Energy</u>	Microscopic	mc^2	Mass



The Universal Rule of the Ledger

Energy is the quantitative property that dictates all motion and interaction in the universe. While its physical forms are endless—from the spin of a diver in mid-air, to the tension of a bent fiberglass pole, to the silent bonds within an atomic nucleus—the fundamental law remains. The currency is never lost, only perfectly traded.